

On the Complexity of Vertex-Splitting Into an Interval Graph

Faisal N. Abu-Khzam^a, Dipayan Chakraborty^{a,b}, Lucas Isenmann^c, Nacim Oijid^{a,d}

^aLebanese American University, Beirut, Lebanon

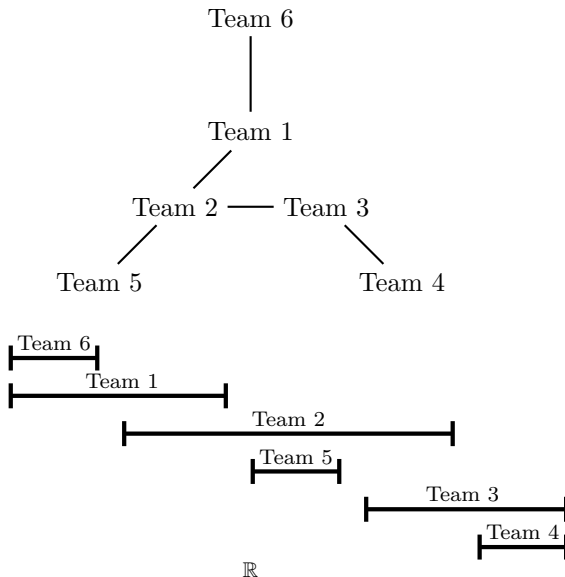
^bCentrale Méditerranée, LIS, Aix-Marseille Université, Marseille, France

^cUniversité de Strasbourg, France

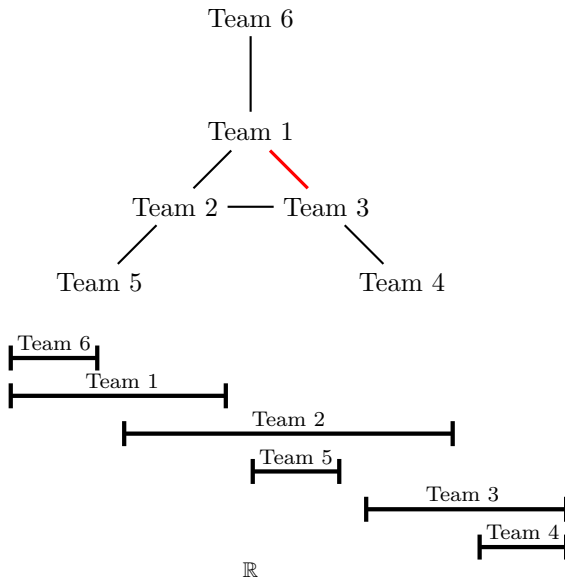
^dUmeå University, Umeå, Sweden

IWOCA 2026, LIMOS, Université Clermont-Auvergne, 11 June 2026

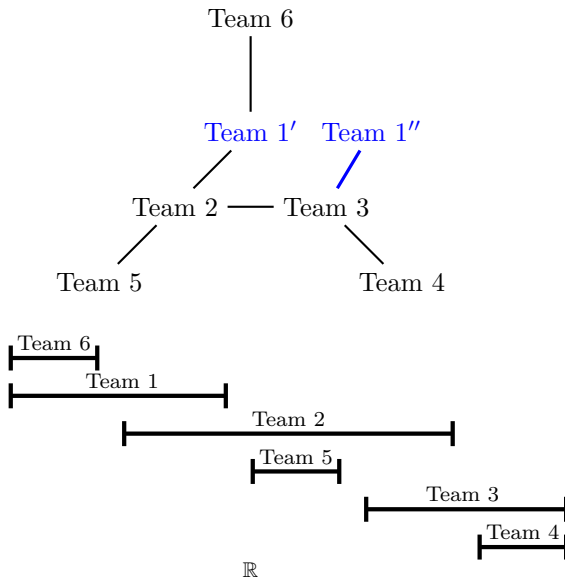
Motivation: Vertex-splitting into Interval Graphs:



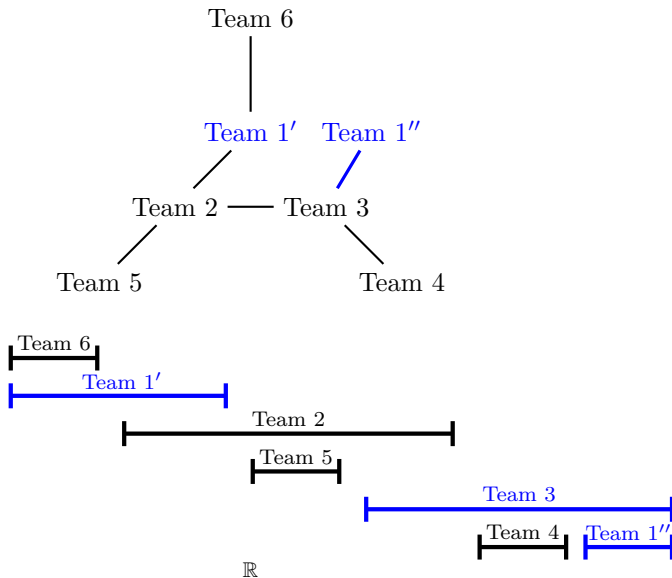
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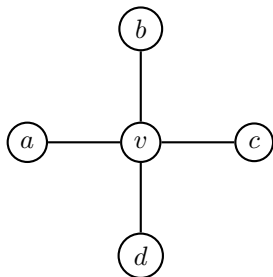
Motivation: Applications of interval graphs

Interval graphs are widely applied to

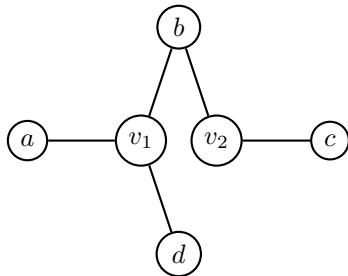
- Scheduling problems
- Resource allocation problems
- Genetics
- Computational biology
- Many otherwise intractable problems become polynomial-time / linear-time on interval graphs

Vertex splitting v :

Replace v by v_1 and v_2 such
that $N(v_1) \cup N(v_2) = N(v)$



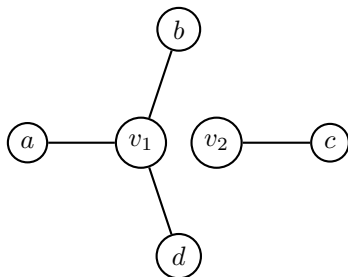
e.g. $\rightarrow$



Exclusive Vertex splitting v :

Replace v by v_1 and v_2 such
that $N(v_1) \cup N(v_2) = N(v)$
and $N(v_1) \cap N(v_2) = \emptyset$

e.g. $\rightarrow$



Vertex splitting was introduced by F. Abu-Khzam, J. Egan, S. Gaspers, A. Shaw and P. Shaw (ISCO 2018).

Π -VERTEX SPLITTING (resp. Π -EX-VERTEX SPLITTING)

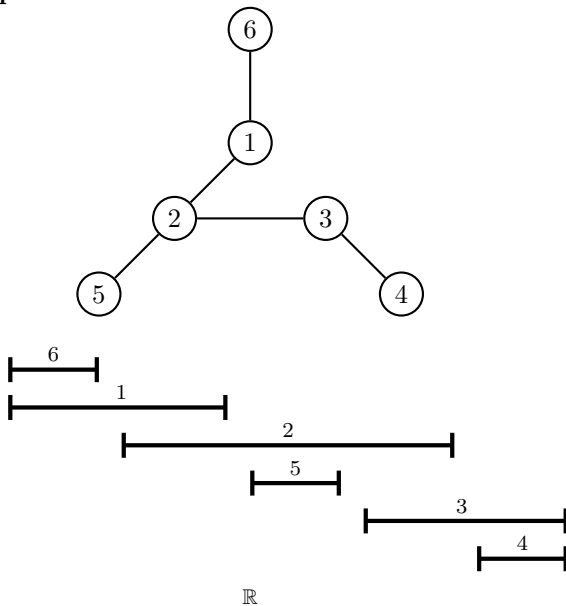
Input: A graph G and a parameter k .

Questions: Does there exist at most k vertex splits (resp. exclusive vertex splits) on G such that the resulting graph belongs to Π

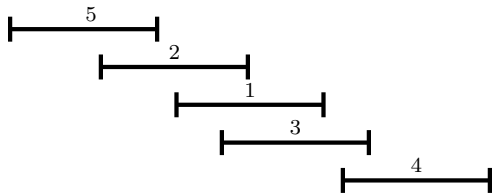
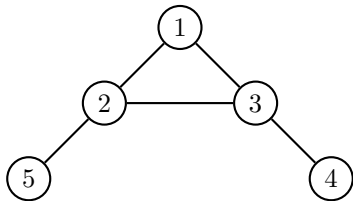
In this work:

$$\Pi = \begin{cases} \text{Interval graphs} \\ \text{Unit Interval Graphs} \\ \text{Chordal Graphs} \end{cases}$$

Interval Graphs:



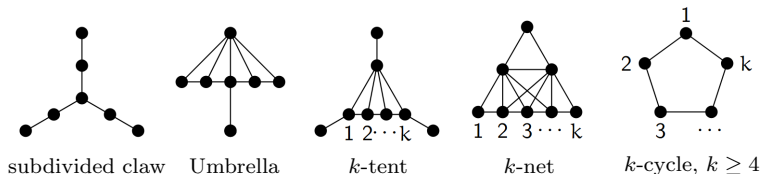
Unit Interval Graphs:



$\mathbb{R}$

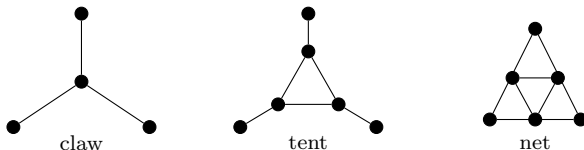
Interval Graphs

Forbidden (induced) subgraph characterization (Lekkerkerker & Boland, 1962)



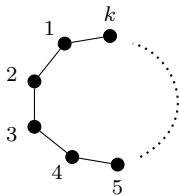
Unit Interval Graphs

Forbidden (induced) subgraph characterization (Wegner, 1967)

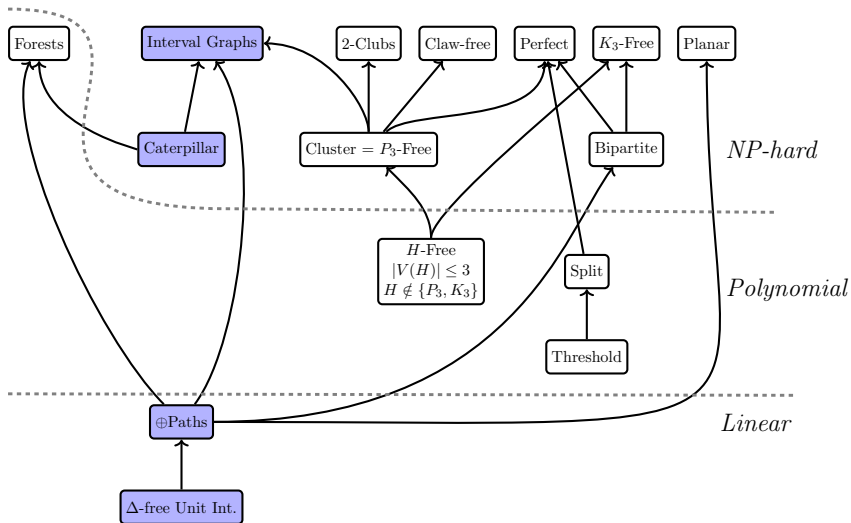


Chordal Graphs

Forbidden (induced) subgraph characterization



k -cycle, $k \geq 4$



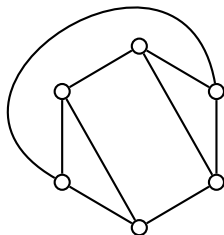
Lemma (Abu-Khzam, C., Isenmann, Oijid, 2026): If the input graph G is **triangle-free**, then

- $IVS(G) = IXVS(G)$
- $UIVS(G) = UIXVS(G)$
- $ChVS(G) = ChXVS(G)$

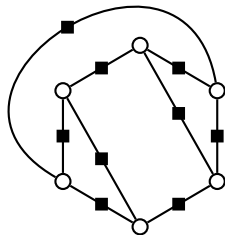
where $\Pi VS(G)$ (resp. ΠXVS) = minimum number of vertex splits (resp. exclusive vertex splits) required such that the resulting graph has property Π .

Theorem (Abu-Khzam, C., Isenmann, Oijid): INT-VERTEX SPLITTING is NP-c, even restricted to planar bipartite subcubic graphs.

Proof: Reduction from [HAMILTONIAN PATH](#). $k = \frac{n}{2} + 1$.
 $|E(G)| = \frac{3n}{2} \implies |E(G')| = 3n$. Also $|V(G')| = n + \frac{3n}{2} = \frac{5n}{2}$.



Planar, cubic G



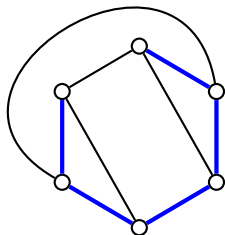
Planar, subcubic, bipartite G'

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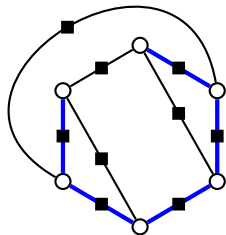
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$(\implies) \#(\text{subdividing vertices not on blue line}) = \frac{3n}{2} - (n - 1) = k$.



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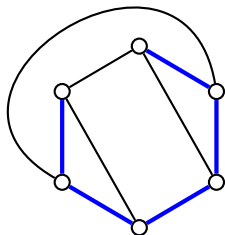
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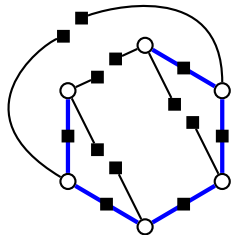
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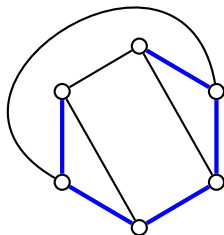
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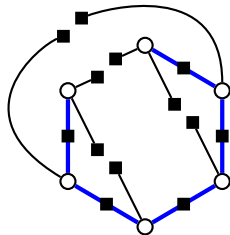
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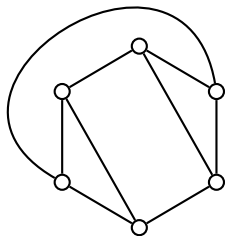
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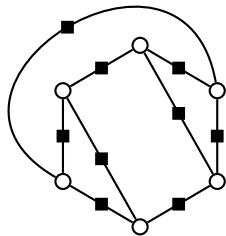
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$(\Longleftarrow)$ Let G' become interval graph T after k splits $\implies T$ is a **forest**.

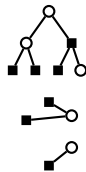


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Planar, subcubic, bipartite G'

VS
 $\rightarrow$



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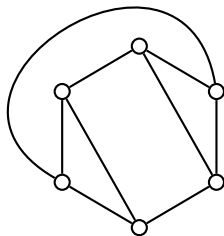
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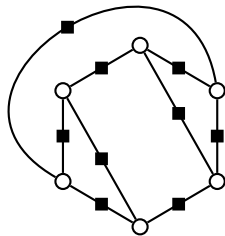
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T is a **tree** since $3n = |E(T)| = |V(T)| - 1 = |V(G')| + k - 1 = 3n$.

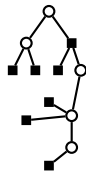


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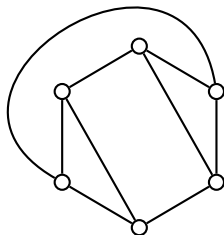
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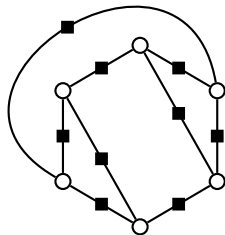
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Subdivided claw-free + tree = caterpillar (Lemma).

Spine of the caterpillar $\implies$ Hamiltonian path of G .

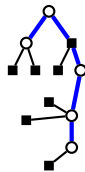


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Corollary: CATERPILLAR-VERTEX SPLITTING is NP-complete, even restricted to planar bipartite subcubic graphs.

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Question: Vertex-splitting in a **Disjoint Union of Paths**?

Theorem (Abu-Khzam, C., Isenmann, Oijid, 2026): Let $G = (V, E)$ be a graph. The minimum number of splits needed to transform G into a **disjoint union of paths** is

$$\sum_{v \in V} \left(\left\lceil \frac{d(v)}{2} \right\rceil - 1 \right) + r,$$

where r is the number of connected components of G that contain no vertex of odd degree.

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Corollary: $\oplus$ PATHS-VERTEX SPLITTING is linear-time solvable.

Corollary: If the input graph is a **triangle-free**, then UNIT
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Proof: Δ -free unit interval graphs $\subseteq$ claw-free forest $= \oplus$ Paths.

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Question: UNIT INT-VERTEX SPLITTING (**in general**)?

The question is still open!

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Problem	Input: Δ -free	Input: general
INT-VERTEX SPLITTING	NP-hard	NP-hard
UNIT INT-VERTEX SPLITTING	Linear	???

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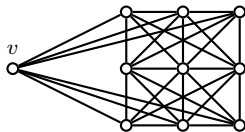
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$ChVS(G) \leq ChED(G)$ but $ChED(G) \not\leq ChVS(G)$ even when $\alpha(G) \leq 2$!

$$ChVS(G) = 1$$

$$ChED(G) = 3$$



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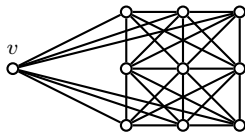
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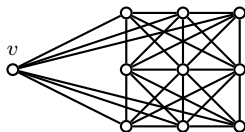
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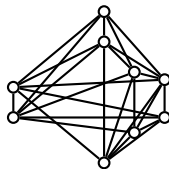


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$ChVD(G) \leq ChVS(G)$ but $ChVS(G) \not\leq ChVD(G)$ even when $\alpha(G) \leq 2$!

$$ChVD(G) = 2$$

$$ChVS(G) \geq 3$$



The picture so far!

Problem	Input: Δ -free	Input: general
INT-VERTEX SPLITTING	NP-hard	NP-hard
UNIT INT-VERTEX SPLITTING	Linear	???
CHORDAL-VERTEX SPLITTING	Polynomial	???

Thank you!